MID-TERM EXAMINATION
(B.Tech CSE AI/ECE AI/ AI&ML) (Semester 1st)
(October, 2023) OFF LINE mode

Subject Code: BAS 109 | **Subject: Applied Mathematics**

Time : 1 ½Hours | **Maximum Marks : 30**

Note: (a) Q. 1 is compulsory. Attempt any two questions from the rest.
(b) Use of calculator is not allowed.

Q1

(a) Define rank of a matrix. Find rank of the matrix **(2.5*4)**

$$A=\begin{bmatrix} 3 & 5 & 2 \\ 4 & 2 & 3 \\ -1 & 2 & 4 \end{bmatrix}.$$

(b) If λ is an eigenvalue of a matrix A then prove that $g(\lambda)$ is an eigenvalue of a matrix $g(A)$, where g is a polynomial.

(c) Find the inverse Laplace transform of $\tan^{-1}(\frac{2}{s^2})$.

(d) Let V be a vector space. Let H and K be subspaces of V. Give an example to show that HUK need not be a subspace.

Q2 **(Attempt any Two Parts) UNIT-1** **(5,5)**

(a) Check whether the system is consistent or inconsistent. Give the complete solution set, if consistent.

$$\begin{aligned} 3x-3y-2z &= 23 \\ -6x+4y+3z &= -40 \\ -2x+y+z &= -12 \end{aligned}$$

(b) Find the inverse of a matrix $A=\begin{bmatrix} 6 & -3 & -2 \\ 4 & -1 & -2 \\ 10 & -5 & -3 \end{bmatrix}$ by using the Cayley-Hamilton theorem.

(c) Consider the matrix:

$$A=\begin{pmatrix} 7 & 1 & -1 \\ -11 & -3 & 2 \\ 18 & 2 & -4 \end{pmatrix}$$

Find the eigenvalues and the fundamental eigenvectors of A.

Q3 **(Attempt any Two Parts) UNIT-2** **(5,5)**

(a) Find the Laplace transform of $f(x)=\begin{cases} 1, & 0\le x\le 1 \\ e^x, & 1<x\le 4. \\ 0, & x>4 \end{cases}$

(b) (i) Consider the vector space $M_{3\cdot 3}(\mathbb{R})$and the sets
T_1: the set of non singular 3*3 matrices
T_2: the set of singular 3*3 matrices.
Determine T_1 and T_2 are subspaces of $M_{3\cdot 3}(\mathbb{R})$ or not?

(ii) Apply convolution theorem to find the inverse Laplace transform of $\frac{1}{(s-2)(s+3)}$.

(c) For the following linear transformation $T:\mathbb{R}^3\to\mathbb{R}^3$

$$T\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 1 & -1 & 5 \\ -2 & 3 & -13 \\ 3 & -3 & 15 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix},$$

find bases for $ker(T)$ and $range(T)$. Also, verify the Dimension theorem.